

DEAKIN UNIVERSITY

FULL STACK DEVELOPMENT: SECURE FRONTEND APPLICATIONS

ONTRACK SUBMISSION

Login and Registration Page

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Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO Unit Learning Outcome 1

September 8, 2026



Login and Registration

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GitLab repository - P4 source files

<https://gitlab.deakin.edu.au/s222528574/task-p4/-/tree/review-p4-20260906/reviewed-app>

Panopto walkthrough - recorded 8 September 2026 (4:18)

<https://deakin.au.panopto.com/Panopto/Pages/Viewer.aspx?id=b15f219a-8630-4b1f-97da-b4bf009ee412>

P4 adds registration and login to DEV@Deakin. React Router connects Home, Login and Sign-up while the header and footer remain shared. The brand links back to Home; Search and Post remain inactive for this stage.

Registration and authentication

Registration collects first name, last name, email, password and confirmation. Browser and Express validation check required fields, email format and matching passwords. The server normalises the email, hashes the password with bcrypt at cost 12 and creates the account atomically in Firestore.

A successful registration returns to Login with the email preserved and an account-created notice, rather than automatically signing in. Login compares the entered password with the stored hash. Correct credentials open Home; incorrect credentials produce a retry message while retaining the sign-up option.

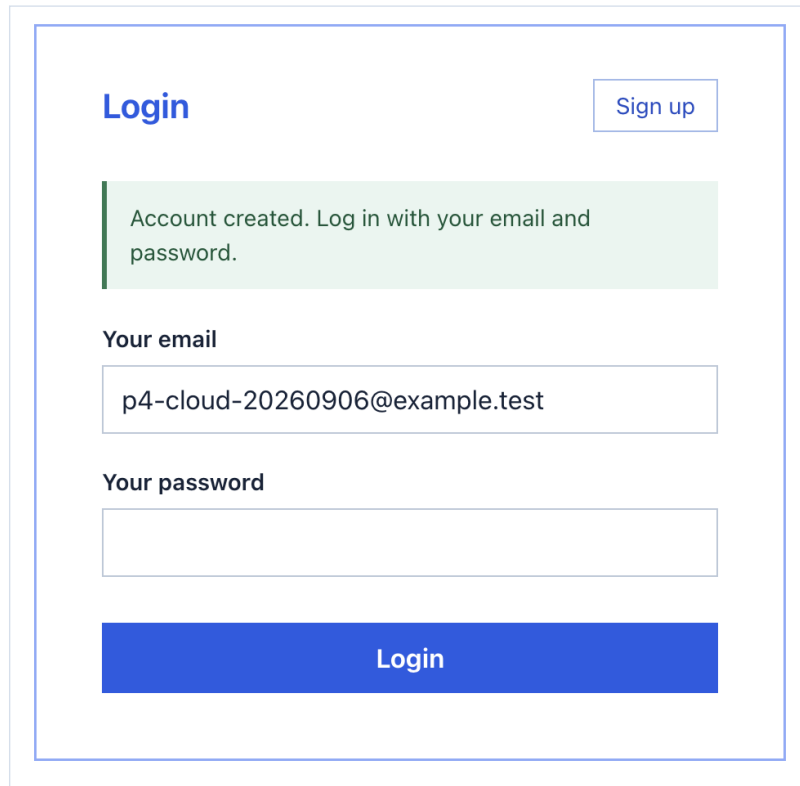
Implementation summary

Area	Implementation
React forms	Controlled input values, field-level errors and pending states. Password limits are eight characters minimum and 72 UTF-8 bytes maximum.
Accounts	p4_users stores accounts. Transactional creation rejects duplicate normalised email addresses without overwriting records.
Sessions	Expiring sessions are kept in p4_sessions. An HttpOnly cookie identifies the session; reload restores the account and logout destroys the session.
Data handling	Plaintext passwords and confirmation are not persisted. Public responses exclude the password hash. Database credentials remain outside the frontend and repository.

Figures 1-6 are browser and Firebase-console captures from the cloud checks on 6 September 2026. They support the separate recorded walkthrough linked above.

Registration and rejected credentials

The cloud test account was created through the browser and returned to the Login page. The blank password field and notice show that registration completed without automatically logging the account in.



The screenshot shows a login interface with a blue border. At the top left is the word "Login" in blue. At the top right is a "Sign up" button. Below this is a green message box that says "Account created. Log in with your email and password." Underneath is a "Your email" label followed by a text input field containing "p4-cloud-20260906@example.test". Below that is a "Your password" label followed by an empty password input field. At the bottom is a large blue "Login" button.

Figure 1. Login-panel detail after creating p4-cloud-20260906@example.test. The local browser and Express service are connected to real Cloud Firestore.

Failure paths

The email or password is incorrect. Try again or create an account.

Figure 2. The incorrect password was rejected. Native browser detail excludes the entered password field.

An account with this email already exists. Log in instead.

Figure 3. Registering the same normalised email again was rejected without replacing the stored account.

Validation and request errors leave the form available for correction. The examples above show actual responses from the local Express application connected to Cloud Firestore.

Session restoration and logout

Correct login returned the same cloud account to the home page. A browser reload then restored Alex Demo from the server-backed session.

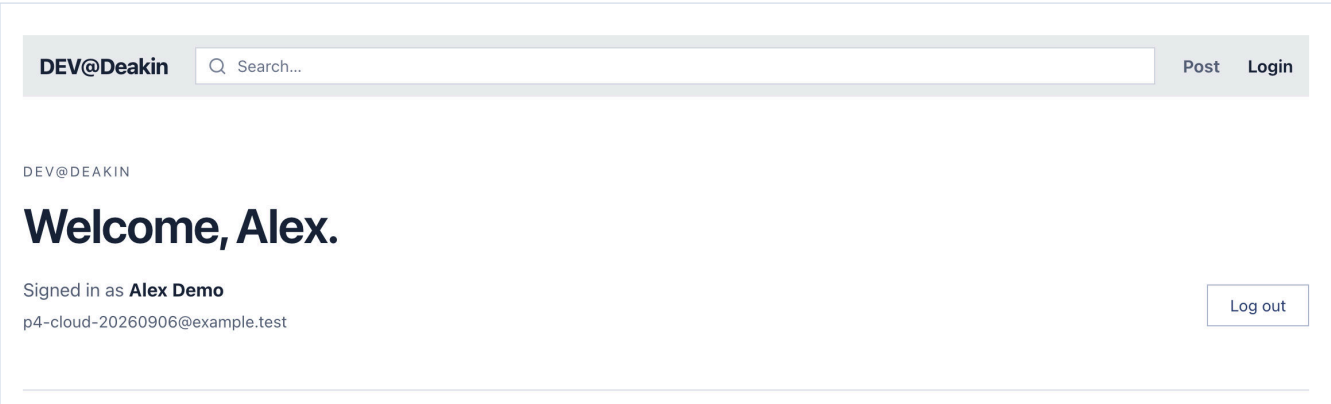


Figure 4. Home-page detail after reload. The account name and email remain visible, with a Log out action.

Logout removed the authenticated account state. The page returned to the public home content, without the account email or logout control.

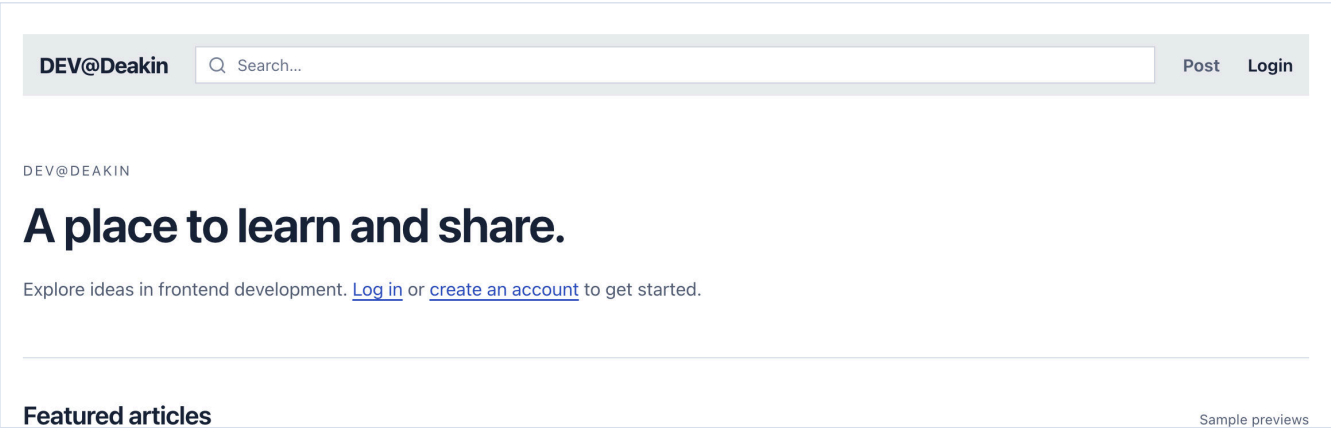


Figure 5. Public home-page detail after logout. This is the real post-logout browser state.

Browser check	Observed result
Register	Redirected to Login with an account-created notice.
Wrong password	Rejected; no authenticated account was shown.
Correct login and reload	Signed-in account restored from the cloud-backed session.
Logout	Authenticated state removed.
Duplicate registration	Rejected; original account retained.

Stored account and verification

The P4 application ran locally at 127.0.0.1:5175 and used real Cloud Firestore in project sit313-p4-1bade, database (default), region australia-southeast2 (Melbourne). The evidence below shows the account saved by the registration flow.

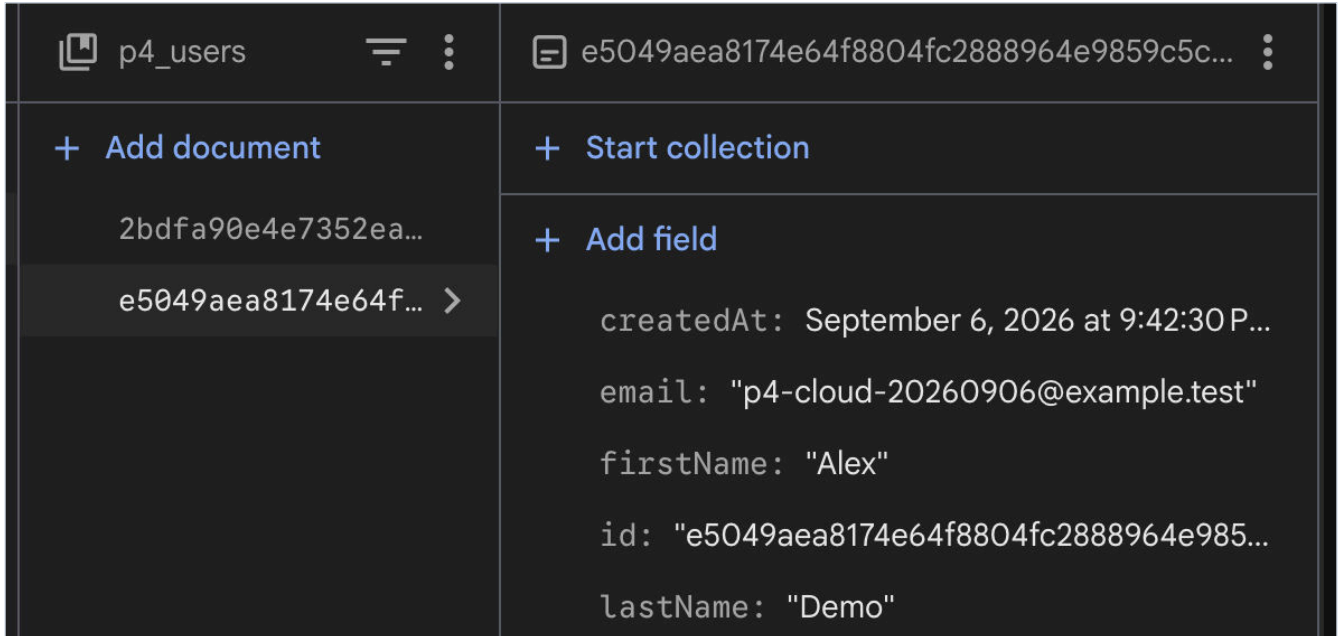


Figure 6. Firebase console account fields from 6 September 2026: email, firstName and lastName match the P4 browser test account. The password hash is outside this view.

Verification results

Check	Result
Browser flow	Registration, incorrect-password rejection, correct login, reload restoration, logout and duplicate-account rejection passed.
Cloud record	A read-only check with the official Firestore SDK confirmed the expected email and names, bcrypt cost 12, a server timestamp and no plaintext-password fields.
Automated tests	36 tests passed: 23 backend/adaptor checks and 13 interface checks.
Published source	The P4 review-branch pipeline passed. The reviewed-app directory contains the task source, resources and tests.
Repository access	Private repository. Aaron Spence and Nisha Kumari were verified as Reporters during preparation.

The recording explains the task interface, React routing, registration and login handlers, and Firestore storage. The screenshots in this report separately document the successful and unsuccessful account operations.